AFS–0633

Sub. Code
7MCE1C1

M.Sc. DEGREE EXAMINATION, APRIL 2018.

First Semester

Computer Science

APPLIED MATHEMATICS FOR COMPUTER SCIENCE

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define tautology. Give example.

2. What is meant by atomic and molecular statements.

3. Define conjunctive normal form. Give example.

4. What is existential quantifies? Give example.

5. What is meant by multigraph? Give example.

6. Define spanning tree.

7. What is canonical form of LPP?

8. Define artificial variables.

9. Define the mathematical formulation of transportation problem.

10. What is meant by unbalanced assignment problem?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the basic connective? Explain with truth table.

Or

(b) Show that :

$$P \to (Q \to R) \Leftrightarrow R \to (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \to R.$$

12. (a) Obtain the conjunctive normal form of $\neg(P \vee Q) \underset{\leftarrow}{\rightarrow} (P \wedge Q)$.

Or

(b) Write short notes on valid formulas and equivalences.

13. (a) Explain matrix representation of graphs in detail.

Or

(b) Define Forest. Explain how an ordered forest can be converted to a unique binary tree.

14. (a) The standard weight of a brick is 5 kg and it contains two basic ingredient B_1 and B_2. B_1 costs Rs.5/Kg and B_2 costs Rs.8/Kg. Strength considerations dictate that the brick contains not more than 4 Kg of B_1 and a minimum of 2 K of B_2. Since the demand for the product is likely to be related to the price of the brick, find graphically the minimum cost of the brick satisfying the conditions.

Or

(b) Use simplex method to solve the following LPP :

Maximize : $Z = 2x_1 + 5x_2$

Subject to $x_1 + 4x_2 \le 24$

$3x_1 + x_2 \le 21$

$x_1 + x_2 \le 9$

$x_1, x_2 \ge 0.$

15. (a) Find the initial basic feasible solution of the following transportation problem by north-west corner rule.

	1	2	3	4	5	Available
A	4	3	1	2	6	80
B	5	2	3	4	5	60
C	3	5	6	3	2	40
D	2	4	4	5	3	20
Required	60	60	30	40	10	200

Or

(b) Explain the Hungarian method for solution of the assignment problem.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. (a) Show that the formula $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ is a tautology.

(b) Write short notes on well-formed formulas.

17. Obtain the principle disjunctive norm and principal conjunctive norm of $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$.

18. Explain rooted trees and binary trees and their representative in detail.

19. Use Two-phase simplex to solve the following LPP.

Maximize : $Z = 5x_1 - 4x_2 + 3x_3$

Subject to $2x_1 + x_2 - 6x_3 = 20$

$6x_1 + 5x_2 + 10x_3 \leq 76$

$8x_1 - 3x_2 + 6x_3 \leq 50$

$x_1, x_2, x_3 \geq 0.$

20. Five wagons are available at stations 1, 2, 3, 4 and 5. They are required at stations I, II, III, IV and V. The mileages between stations are given below. How should the wagons be assigned so as to minimize the total mileage covered?

	I	II	III	IV	V
1	10	5	9	18	11
2	13	9	6	12	14
3	3	2	4	4	5
4	18	9	12	17	15
5	11	6	14	19	10

AFS–0634

Sub. Code
7MCE1C2

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

DESIGN AND ANALYSIS OF ALGORITHMS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** the questions.

1. Define space and time complexity of an algorithm.

2. Define a binary tree.

3. Differentiate defective chessboards with that of $k = 0$ and $k > 0$.

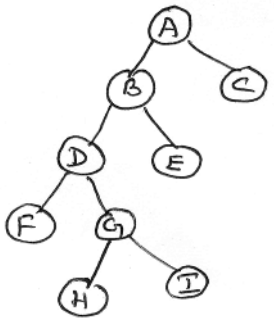
4. Compare quick sort with merge sort.

5. Define a feasible solution.

6. Draw an undirected graphs with 4 nodes with three of its spanning trees.

7. Define the principle of optimality.

8. Give the in order transversal of the binary tree.



9. Define an implicit constraint.

10. When does a graph is said to be a planar?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write briefly about the three popular methods to arrive at amortized casts for operations.

Or

(b) Describe the operations performed on a stack with its algorithm.

12. (a) How can you find the maximum and minimum recursively? Discuss with its algorithm.

Or

(b) Prove that the average computation time TA(n) of the function select 1 is O(n).

13. (a) Write in short about container loading.

Or

(b) How can you find shortest paths from a single source? Explain with an example.

14. (a) Construct the possible binary search trees for the identifier set {do, if, while}.

Or

(b) Discuss about string editing with an example.

15. (a) Let $w = \{5, 7, 10, 12, 15, 18, 20\}$ and m = 35. Find all possible subsets of w that sum to m.

Or

(b) Describe the 15 puzzle arrangement problem with its goal arrangement.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about the commonly used representations for graphs with examples.

17. Explain the merge sorting procedure and analyse the same by taking an example.

18. Describe the working of Prim's Algorithm by taking an example tree.

19. Explain the travelling salesman problem and analyse the same.

20. Discuss in detail about graph coloring by taking an example graph.

AFS–0635

Sub. Code
7MCE1C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

ADVANCED JAVA PROGRAMMING

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the three main types of statements that the JDBC supports?

2. Write the use of connection class.

3. What is the use of URL connection?

4. What is the use of server socket class?

5. Define introspection.

6. What is BDK?

7. Which method of servlet class called when the servlet is initialized?

8. Name the packages that contain classes and interfaces to build servlets.

9. Name any two AWT classes.

10. Write the properties of button control.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the anatomy of JDBC application.

Or

(b) Write a note on SQL exceptions.

12. (a) Write a note on TCP/IP server sockets.

Or

(b) Give a suitable example program for networked communications client and server.

13. (a) How to create and configure an instance of the our button bean?

Or

(b) Write a note on application builder tools.

14. (a) Explain servlet interface.

Or

(b) Describe session tracking.

15. (a) Explain combo box with example.

Or

(b) Give suitable example to demonstrate the color using applet.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a detailed note on metadata functions.

17. Give a brief note on TCP/IP client sockets.

18. Discuss about introspection.

19. Discuss the methods in Http servlet class.

20. Explain the methods in graphics class with example.

AFS–0636

Sub. Code
7MCE1C4

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

PRINCIPLES OF COMPILER DESIGN

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. What is a translator?

2. Define NFA.

3. When will the grammar is said to be ambiguous?

4. Draw a parse tree for $5 + 3 * 2$.

5. Write down the postfix form of $(a + b) * (c + d)$.

6. Specify the general form of three-address code.

7. Give any two block-structured languages.

8. List down any two syntax errors.

9. Define a basic block.

10. What is a DAG?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Write a brief about the principal aids provided by compiler – compilers.

Or

(b) Describe about input buffering.

12. (a) Construct a parse tree for $id + id * id$ with its derivation.

Or

(b) Describe LR parser with its schematic representation.

13. (a) How can you evaluate the postfix expression? Explain with an example.

Or

(b) Discuss in brief about mixed mode expressions.

14. (a) Write about self-organising lists.

Or

(b) Discuss in brief the errors seen by each phase of a compiler.

15. (a) Write an algorithm to partition into Basic blocks.

Or

(b) Discuss about the various addressing modes used.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe the structure of a compiler in detail.

17. Discuss about shift-reduce parsing with its stock implementation.

18. Explain the methods of translating Boolean expressions.

19. How will be the storage allowcation is done in FORTRAN? Explain.

20. Discuss about Peephole optimization.

AFS–0637

Sub. Code
7MCE1E1

M.Sc. DEGREE EXAMINATION, APRIL 2018.

First Semester

Computer Science

***Elective* — OBJECT ORIENTED ANALYSIS AND DESIGN**

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What are the three kinds of models to describe a system?

2. Differentiate aggregation and associations.

3. State the purpose of state diagram.

4. Define DFD.

5. Write the steps to be performed in constructing an object model.

6. What is association?

7. What is the use of data store?

8. List out the limitations of current DBMS.

9. What is the purpose of object design phase?

10. What is meant by two-way associations?

Section B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Describe the different forms of associations and links with example.

Or

(b) State the reasons for overriding and write the semantic rules for inheritance.

12. (a) Draw a state diagram for that can represent phone line with actions and activities.

Or

(b) Briefly discuss about synchronization of concurrent activities.

13. (a) Discuss the steps to keep the right associations.

Or

(b) How will you refine the classes using inheritance?

14. (a) Write the difference between procedure driven system and event driven system.

Or

(b) Give a detailed note on transaction manager.

15. (a) How will you combine the object, dynamic and functional models?

Or

(b) Write short notes on state machine engine.

Section C

(3 × 10 = 30)

Answer any **three** questions.

16. Prepare an object model to describe the workstation of window management system and explain.

17. Give a brief note on concurrency with suitable example.

18. Explain the steps involved in constructing a dynamic model.

19. Explain how to allocate the subsystem to preprocessors and tasks.

20. Discuss the design of association in object design.

AFS–0638

Sub. Code
7MCE1E2

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

Elective — SYSTEM SOFTWARE

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Specify the two language processing activities.

2. What is hybrid entry format? How it is depicted?

3. Define DFA.

4. What is the purpose of DC statements?

5. How can you call a macro?

6. What is an expression tree?

7. Define linking.

8. What are overlays?

9. Specify the function of a relocating loader.

10. Mention the two components of UI.

Section B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) How grammers can be classified? Explain with its characteristics.

Or

(b) Discuss about binary search organisation.

12. (a) Build a DFA which can recognize integers, real numbers and identifiers.

Or

(b) Discuss about the advanced assembler directives with an example.

13. (a) Write in brief about nested macro calls with examples.

Or

(b) Differentiate static and dynamic memory allocation in brief.

14. (a) Differentiate pure and impure interpreters with its diagrammatic representation.

Or

(b) Write briefly about object module.

15. (a) Discuss about the software tools used in program testing and debugging.

Or

(b) List down the facilities provided by the debug monitors for dynamic debugging.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about the two allocation data structures in brief.

17. Describe the design specification of an assembler with its data structures

18. Explain same of the optimizing transformations used in compilers with examples.

19. Explain the design of a linker for MS-DOS in brief.

20. Discuss in brief about user interfaces.

AFS–0639

Sub. Code
7MCE1E3

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Computer Science

***Elective* — SOFTWARE ENGINEERING**

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is meant by ubiquitous computing?

2. State any two drawbacks of RAD model.

3. What is use-case?

4. Define analysis patterns.

5. State why developing an estimation approach with use-cases is problematic?

6. What is COCOMO?

7. Distinguish between top-down integration and bottom-up integration.

8. Define Recovery testing.

9. What are the direct measures of software process?

10. What is the intent of Domain Engineering?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss about the characteristics of a software.

Or

(b) Explain Waterfall model in detail.

12. (a) Describe the steps required to initiate requirements engineering process.

Or

(b) Explain how to develop effective use-cases.

13. (a) Write short notes on COCOMO II model.

Or

(b) Describe the approach used for the estimation of object oriented projects.

14. (a) Write short notes on Smoke testing.

Or

(b) Discuss in brief about the types of system tests for software-based systems.

15. (a) State the measures of software quality that provide useful indicators of the project team. Explain.

Or

(b) Write short notes on CBSE process.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe in detail about the incremental process models.

17. Write in detail about eliciting requirements.

18. Describe in detail about the techniques used to estimate the project with examples.

19. Explain in detail about black box testing.

20. Discuss in detail about component-based development.

AFS–0640

Sub. Code
7MCE2C1

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

COMPUTER SYSTEM ARCHITECTURE

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is the role of arithmetic logic unit?

2. What are the three phases of an instruction cycle?

3. What is meant by micro operation?

4. Define instruction code.

5. Define control word.

6. What is the purpose of micro program sequencer?

7. What is meant by input-output interface?

8. Expand UART. Define bandrate.

9. Define throughput.

10. What is hardware interlocks?

Section B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short notes on base register addressing mode.

Or

(b) Explain briefly about the types of interrupts.

12. (a) Design a 4-bit arithmetic circuit and explain it.

Or

(b) Write short notes on timing control.

13. (a) Write briefly about conditional branching.

Or

(b) Write short notes on symbolic micro instructions.

14. (a) Write about I/O versus memory bus.

Or

(b) Write short notes on handshaking.

15. (a) Explain arithmetic pipeline briefly.

Or

(b) Write short notes on SIMD array processor.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain data manipulation instructions in detail.

17. Discuss the arithmetic, logic and shift micro operation with examples.

18. Discuss the micro program sequencer.

19. Explain input-output processor with neat diagram.

20. Explain RISC architecture.

AFS–0641

Sub. Code
7MCE2C2

M.Sc. DEGREE EXAMINATION, APRIL 2018.

Second Semester

Computer Science

.NET TECHNOLOGY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is CLR?

2. Define polymorphism.

3. List out the data types in VB.NET.

4. Write the syntax of Inputbox.

5. List out the properties of list box.

6. What is the use of status bar.

7. State the purpose of web-config.

8. Write three key features of HTML server controls.

9. What is authentication?

10. What is the use of dataset?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write a note on assemblies.

Or

(b) Explain inheritance and overloading.

12. (a) Discuss about conditional statement with example.

Or

(b) Describe rich text box control.

13. (a) Give a suitable example for button and checkbox.

Or

(b) Discuss about toolbars with example.

14. (a) Explain checkbox and radiobutton control with example.

Or

(b) Give a suitable example for Rangevalidator control.

15. (a) Discuss the procedure for ASP.NET forms authentication.

Or

(b) Write a note on connection string.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about .Net component.

17. Describe MDI form with example.

18. How to design a menu? Give example.

19. Explain session state with example.

20. Explain the types of SQL statement with example.

AFS–0642

Sub. Code
7MCE2C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

DISTRIBUTED OPERATING SYSTEM

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is access transparency?

2. Write the two commonly used LAN topologies.

3. List the main reliability issues in designing a message passing system.

4. What is meant by "Ordered message delivery"?

5. What is processor consistency model?

6. Define Knot.

7. Write the main purposes of using files.

8. What is cache location?

9. Define Encryption.

10. List the goals of computer security.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) List and explain the DCE components.

Or

(b) Discuss the main features of ATM technology.

12. (a) Describe about three different process addressing mechanisms.

Or

(b) Discuss about blocking and non-blocking types of IPC.

13. (a) Discuss about design and implementation issues of DSM.

Or

(b) Explain the centralized approach for deadlock detection.

14. (a) Explain about file models.

Or

(b) Write the advantages of replication.

15. (a) What are the commonly used approaches for user authentication in computer system? Explain.

Or

(b) Explain the password-based approach for user logins authentication.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the issues in designing a distributed operating system.

17. List and explain any two group communication.

18. Explain the event ordering in distributed system.

19. Explain the most flexible concurrency control algorithm.

20. Discuss about the commonly used methods for implementing an access matrix.

AFS–0643

Sub. Code
7MCE2E1

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

***Elective* — MOBILE COMPUTING**

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Mobile IP?

2. What is Mobility?

3. Define Signals.

4. Write the four different types of multiplexing.

5. What is authentication?

6. What is Mobile node?

7. Define Encapsulation.

8. What is datagram?

9. What is DHCP?

10. Define WAP.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write a note on wireless technologies.

Or

(b) Give an overview of protocol.

12. (a) Describe the services of Antennas.

Or

(b) What are the advantages and disadvantages of cellular systems?

13. (a) Explain Mobility Agent operation.

Or

(b) Discuss about Registration procedure for Home agent.

14. (a) Explain Unicast datagram routing.

Or

(b) Discuss about route optimization message formats.

15. (a) Explain the binding options of IP.

Or

(b) Give a brief note on reverse tunnelling.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about the overview of IP and Routing.

17. Explain the methods for translating digital to analog signal.

18. Describe overview of Authentication.

19. Explain tunnel management.

20. Explain how DHCP is used for mobility and portability.

AFS–0644

Sub. Code
7MCE2E2

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

Elective : GRID COMPUTING

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Grid?

2. Write any two features of I-WAY (SC95).

3. Write down any four real and specific problem that underlies grid concept.

4. Why are services important?

5. List the advantages of digital libraries.

6. What is a content link?

7. Define : OGSI specification.

8. What is the use of service entries in a service group?

9. State the difficulties in resource management.

10. What are the responsibilities of resource owner?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the community grid model with neat diagram.

Or

(b) Briefly write the evolution of the grid.

12. (a) Explain the emergence of virtual organization.

Or

(b) Write short notes on semantic grids.

13. (a) How to enable network services in grid? Explain.

Or

(b) What are the key properties of web service discovery architecture?

14. (a) Explain about service data structure, elements and values in OGSI specification.

Or

(b) Explain the service domains of OGSA.

15. (a) Explain how the brain activity analysis grid application is used.

Or

(b) Write short note on scheduling in grid applications.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the scientific applications of grid.

17. Describe the architecture of grid.

18. Explain the data virtualization services in detail.

19. Discuss about naming and change management recommendations of grid services in detail.

20. Explain the deployment of grid applications with a sample scenario.

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AFS–0645

Sub. Code
7MCE2E3

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

Elective : COMPUTER GRAPHICS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Resolution.

2. What is Scan conversion?

3. State the transformation matrix for Rotation.

4. What is meant by composite transformation?

5. Distinguish between window and viewport.

6. What is the purpose of graphical input functions?

7. What is perspective projection?

8. What is the purpose of exploding an object?

9. Define axonometric orthographic projection.

10. What is vanishing point?

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) Describe DDA line drawing algorithm in detail.

Or

(b) Write short notes on Raster and Random Scan Systems.

12. (a) Explain the Reflection and Shear Transformations in 2D.

Or

(b) Explain general pivot-point rotation and general fixed-point scaling.

13. (a) Describe Cohen-Sutherland line clipping algorithm in detail.

Or

(b) Write briefly about various Input devices.

14. (a) Explain the three dimensional basic transformations in detail.

Or

(b) Discuss briefly about the three-dimensional display techniques.

15. (a) Explain scan line method in detail.

Or

(b) Write briefly about projections.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss briefly about Video Display Devices

17. Explain in detail about the following :

    (a) Line attributes

    (b) Character attributes

    (c) Bundled attributes.

18. Enlighten about the logical classification of input devices.

19. Explain three-dimensional viewing in detail.

20. Explain Depth buffer and scan line methods in detail.

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AFS–0646

Sub. Code
7MCE2E4

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

Elective – PARALLEL PROCESSING

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define parallel processing.

2. What is pipelining?

3. What is loosely coupled systems?

4. What is linear network?

5. Define data parallelism.

6. Define message passing.

7. What is Granularity?

8. Write any two performance laws.

9. List out three shared memory multiprocessor models.

10. What is shared bus?

Part B $(5 \times 5 = 25)$

Answer **all** the questions, choosing either (a) or (b).

11. (a) What are the requirements to implement parallel processing?

Or

(b) Describe the memory related issues in parallel processing.

12. (a) Write a note on tightly coupled systems.

Or

(b) Discuss about two dimensional mesh and hypercube.

13. (a) Differentiate data and control parallelism.

Or

(b) Describe the types of Granularity.

14. (a) Explain the design issues of parallel algorithm.

Or

(b) Discuss about branch and bound search algorithm.

15. (a) Write a note on cross bar and multiport memory.

Or

(b) Explain how to handle shared memory.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss the mechanisms of implementing parallel processing.

17. Give a brief note on linear and Ring Networks.

18. Explain Precedence Graph of process.

19. Describe any two performance measures and analysis.

20. Explain in detail about cache Coherence.

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AFS–0647

Sub. Code
7MCE2E5

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

Elective — ADVANCED DATABASE SYSTEMS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a database?

2. Define data independence.

3. Define primary key.

4. Write the purpose of RENAME operator in relational Algebra.

5. What do you mean by functional dependency?

6. Why is it necessary to decompose a relation?

7. What is replication?

8. What is location transparency?

9. What is hybrid peer-to-pear network?

10. List the emerging database technologies.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What is data dictionary? Explain.

Or

(b) Discuss the role of database administrator.

12. (a) List the unary operations in relational Algebra. Explain.

Or

(b) Write a note on E-R diagram.

13. (a) Discuss about the fourth normal form with an example.

Or

(b) Why Boyce codd normal form is considered a stronger Form of 3NF? Explain.

14. (a) Why are distributed databases desirable? Explain.

Or

(b) Discuss about two-phase commit protocol.

15. (a) What is an infrastructure less mobile platform? Explain.

Or

(b) Discuss about the nature of multimedia data.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. List the types of data models and explain.

17. Explain the various set operation of relational algebra with an example.

18. Explain insertion, deletion and modification anomalies with an example.

19. Explain about the concurrency control.

20. Write a note on MySQL databases.

AFS–0648

Sub. Code
7MCE2E6

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Computer Science

Elective — DIGITAL IMAGE PROCESSING

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is digital image processing?

2. Define image sampling and quantization.

3. Define gamma correction.

4. What is meant by contrast stretching?

5. Define power spectrum.

6. Define Gaussian Highpass filter.

7. Define salt-and-pepper noise.

8. What is meant by blind deconvolution?

9. Define radiance, luminance and brightness.

10. What is meant color slicing?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the principal modalities used for image sensing and acquisition.

Or

(b) Write short notes on Gamma-Ray imaging and X-Ray imaging.

12. (a) Write short notes on histogram processing.

Or

(b) Explain the basic five steps used to implement a fuzzy-rule based system.

13. (a) Explain Image Smoothing using frequency domain filters in detail.

Or

(b) Write briefly about Selective filtering.

14. (a) Write short notes on Noise Models.

Or

(b) Explain in detail about the model of the image degradation/restoration process.

15. (a) Explain the process of image segmentation based on color.

Or

(b) Write briefly about digital image watermarking.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about the relationships between pixels in a digital image.

17. Describe the basic intensity transformation functions in detail.

18. Enlighten the properties of the 2-D Discrete Fourier Transform.

19. Explain the various ways to estimate the degradation function.

20. Describe in detail about any four basic image compression methods.

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